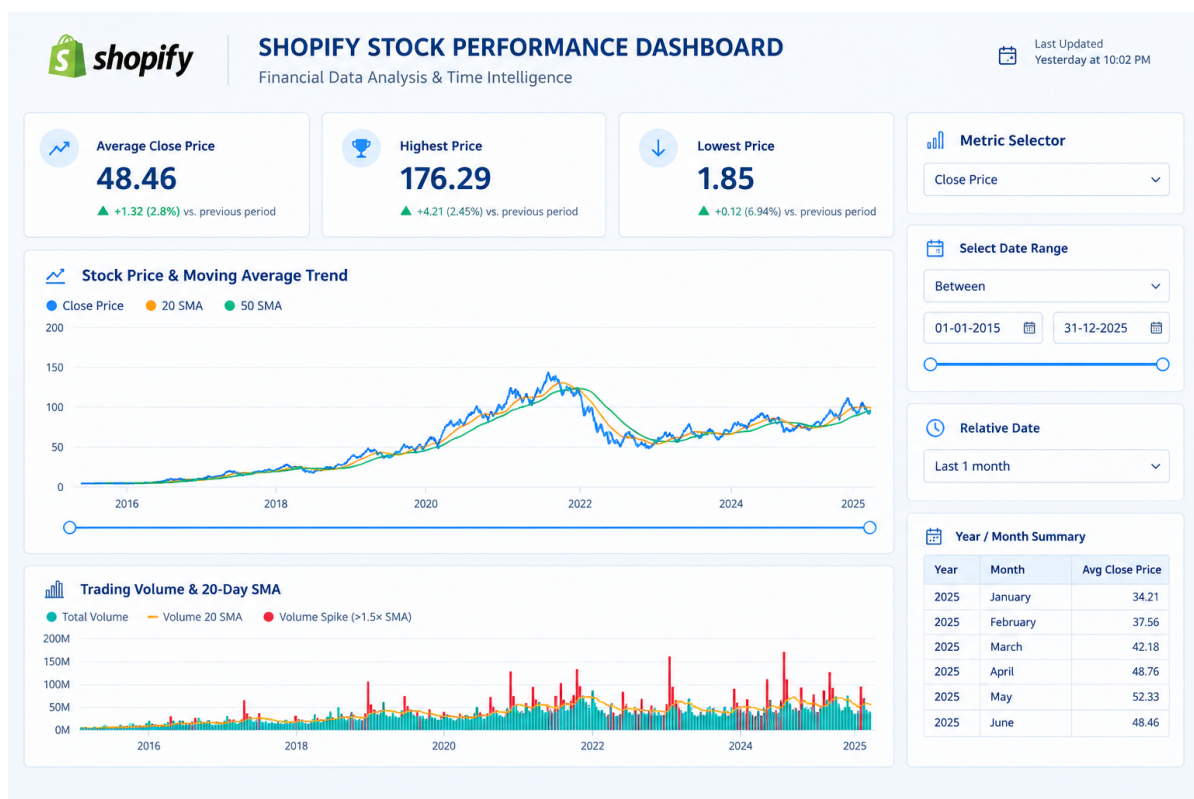


SHOPIFY STOCK ANALYSIS

Week 4 – Power BI Exploratory Data Analysis

Moving Averages • Crossover Signal • Interactive Price & Volume Analysis • Dynamic Slicers
• Tooltip Page



Final dashboard layout reference used to organize the Week 4 Power BI report.

Project	Shopify Stock Performance Dashboard
Tool	Microsoft Power BI Desktop
Week	Week 4
Continuation	Built as a continuation of the Week 3 Power BI project
Primary dataset	Shopify_Stock with Dim_Date

1. Project Overview

The Week 4 work extends the Week 3 Shopify stock dashboard by adding time-series moving averages, a moving-average crossover indicator, interactive price and volume visuals, date controls, a metric selector, and a report-page tooltip. The task document specifies 20-day and 50-day SMA measures, continuous time axes, volume-spike highlighting, relative date filtering, field parameters, and a Stock_Tooltip page.

Week 4 objectives

#	Requirement	Implementation
1	20-day & 50-day SMA	DAX measures using DATESINPERIOD
2	Crossover signal	20 SMA vs 50 SMA → Bullish/Bearish text
3	Interactive price trend	Close Price + 20 SMA + 50 SMA + continuous date + zoom
4	Volume/volatility	Total Volume + 20 SMA + spike color formatting
5	Timeline controls	Between Date Range + Relative Date slicer
6	Metric Selector	Open / High / Low / Close field parameter
7	Tooltip	Stock_Tooltip with six detail metrics

Week 3 continuation

The existing Week 3 KPI cards, date range control and summary table can remain. The Week 4 visuals should update the main price chart and add the dedicated volume analysis rather than deleting the whole Week 3 dashboard.

2. DAX – Moving Average Measures

20-Day Volume Simple Moving Average

```
Volume 20 SMA =  
VAR CurrentDate = MAX(Dim_Date[Date])  
VAR PeriodDays = 20  
VAR DateInPeriod =  
    DATESINPERIOD(  
        Dim_Date[Date],  
        CurrentDate,  
        -PeriodDays,  
        DAY  
    )  
RETURN  
    CALCULATE(  
        AVERAGE(Shopify_Stock[Volume]),  
        DateInPeriod  
    )
```

50-Day Volume Simple Moving Average

```
Volume 50 SMA =  
VAR CurrentDate = MAX(Dim_Date[Date])  
VAR PeriodDays = 50  
VAR DateInPeriod =  
    DATESINPERIOD(  
        Dim_Date[Date],  
        CurrentDate,  
        -PeriodDays,  
        DAY  
    )  
RETURN  
    CALCULATE(  
        AVERAGE(Shopify_Stock[Volume]),  
        DateInPeriod  
    )
```

Moving Average Crossover Signal

```
MA Crossover Signal =  
IF(  
    ISBLANK([20 SMA]) ||  
    ISBLANK([50 SMA]),  
    BLANK(),  
    IF(  
        [20 SMA] > [50 SMA],  
        "Bullish (20 > 50)",  
        "Bearish (20 < 50)"  
    )  
)
```

Note: The task document uses the names [20 SMA] and [50 SMA] inside the crossover expression, while the supplied measure headings are "Volume 20 SMA" and "Volume 50 SMA". In Power BI, use the exact names of the measures you created consistently in the final expression.

3. Interactive Price Trend Chart

The main Week 4 chart should show the stock close price together with the 20-day and 50-day moving-average lines. The date axis must be continuous so the time series is displayed as an unbroken timeline, and the Zoom slider provides quick timeline navigation.

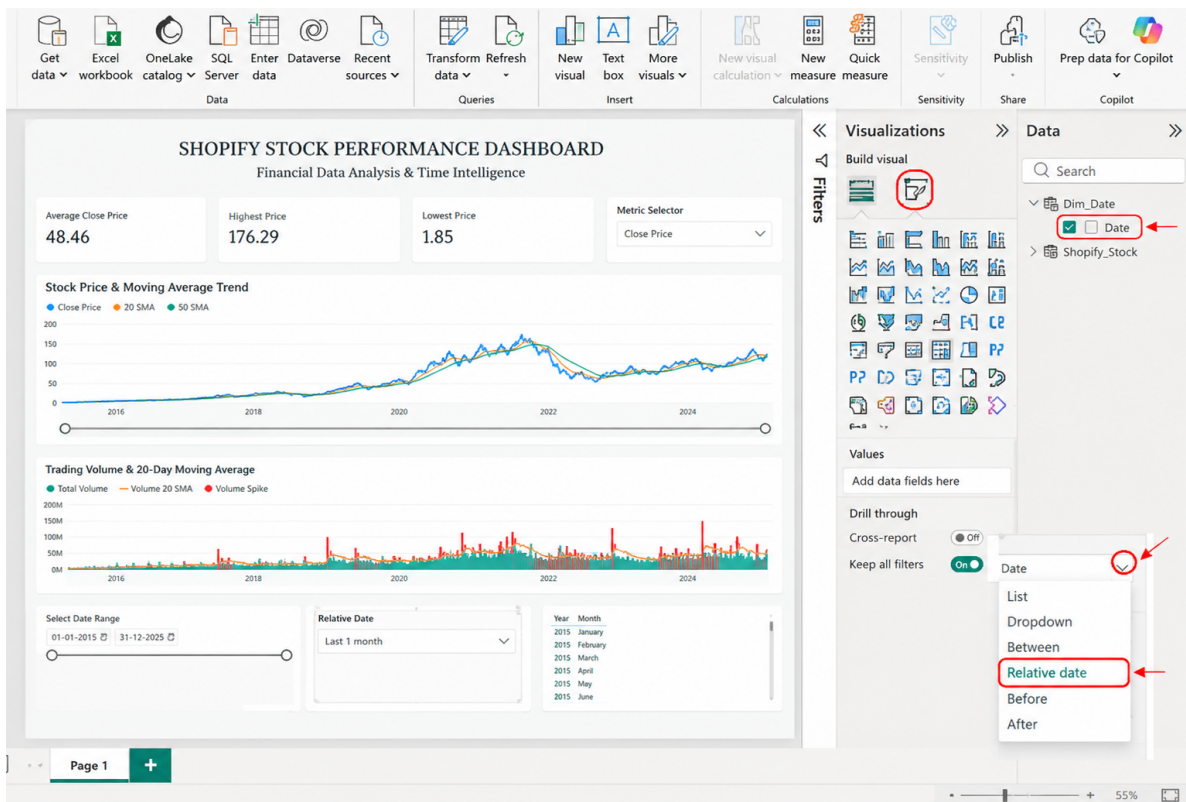


Figure 1. Power BI dashboard screenshot showing the price chart, metric selector, date controls and relative-date setup.

Recommended field setup

Chart area	Field / setting
X-axis	Dim_Date[Date]
Line values	Close Price, 20 SMA, 50 SMA
X-axis type	Continuous
Zoom slider	On
Legend	On – identifies Close Price / 20 SMA / 50 SMA

4. Trading Volume & Volatility Visual

A separate visual is placed below the price chart. It contains Total Volume and the 20-day SMA. The task requires volume spikes above 1.5× the 20-day SMA to be highlighted differently from normal volume.

Conditional color measure

```
Volume Color =  
IF(  
[Total Volume] > [20 SMA] * 1.5,  
"#FF4D4D",  
"#26A69A"  
)
```

Apply the Volume Color measure through the visual's Columns → conditional formatting (FX) so that spike bars use the spike color and normal bars use the normal volume color.

Dashboard screenshot showing the volume visual

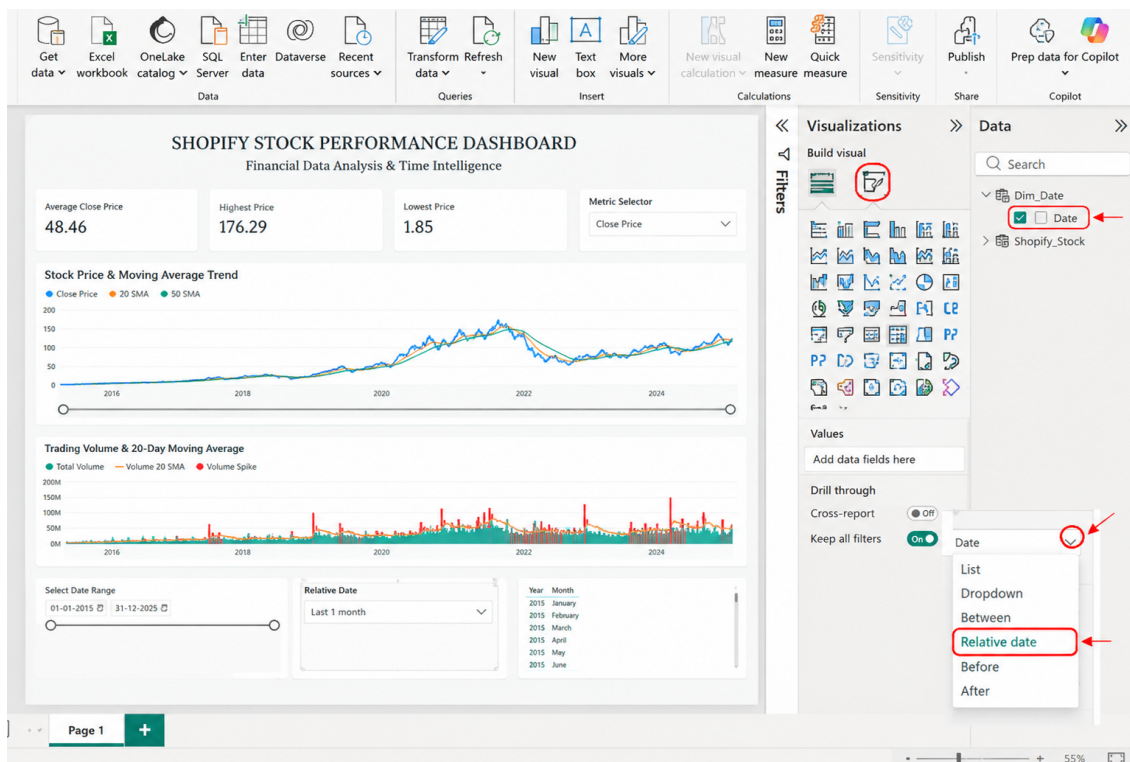


Figure 2. Week 4 dashboard working view with the volume analysis area below the price trend.

5. Timeline Slicers & Metric Selector

Date Range Slicer

Keep the existing Date Range slicer and set its mode to Between. This provides a manual start-date/end-date range control for the dashboard.

Relative Date Slicer

Add a second slicer using Dim_Date[Date] and change the slicer type to Relative date. The task document specifies quick intervals of 1 month, 3 months, 6 months, YTD and 1 year. Power BI's standard relative-date control may require the interval value/unit to be changed; if the five choices must appear together as a single user-facing selector, use a field-parameter/selector design rather than claiming the standard slicer itself supplies five fixed buttons.

Metric Selector

Create a field parameter from Modeling → New parameter → Fields and add Close Price, High Price, Low Price and Open Price. Place the generated parameter in a slicer so the user can switch the metric displayed by the selected visual.

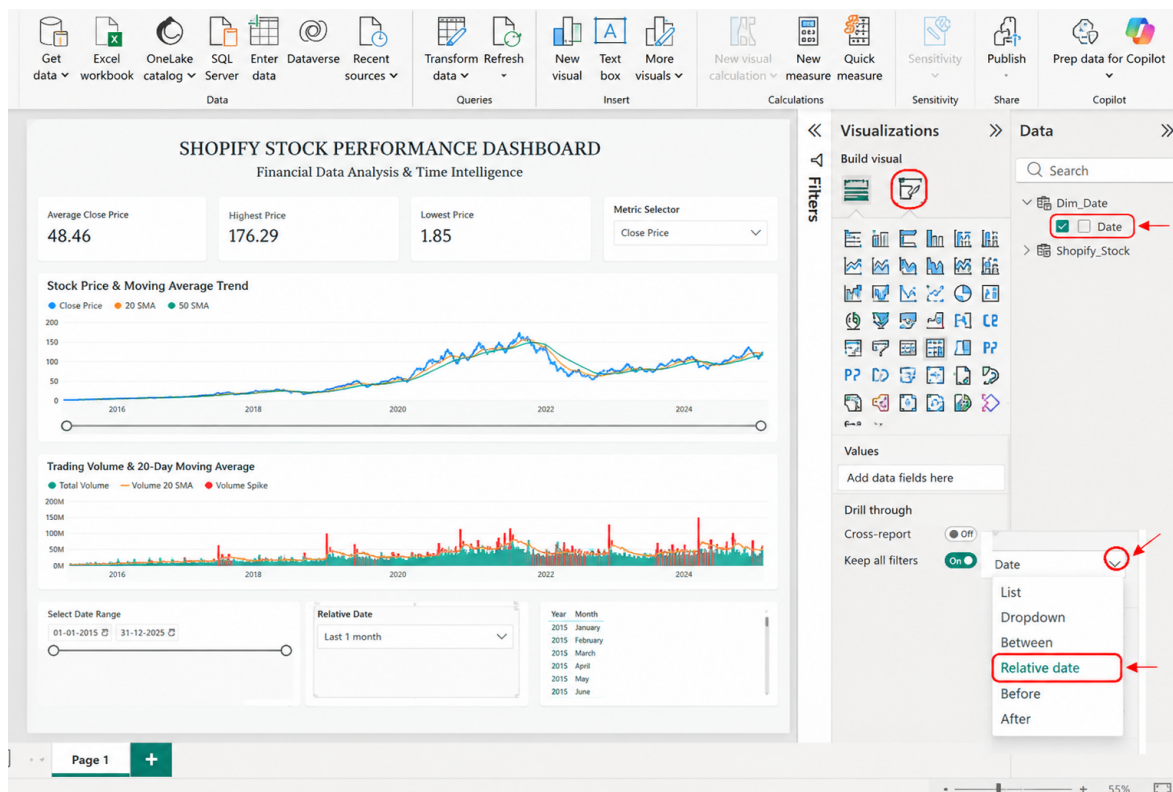


Figure 3. Screenshot used as the visual guide for Relative Date and Metric Selector placement.

6. Custom Tooltip Page

The tooltip page is a small report page that appears when the user hovers over the main chart. The task requires the page to be named Stock_Tooltip, enabled as a tooltip page, and populated with selected-date metrics.

Required tooltip metrics

Metric	Power BI field / measure
Open	Open Price
High	High Price
Low	Low Price
Close	Close Price
Volume	Total Volume
Daily Change %	Daily Change %

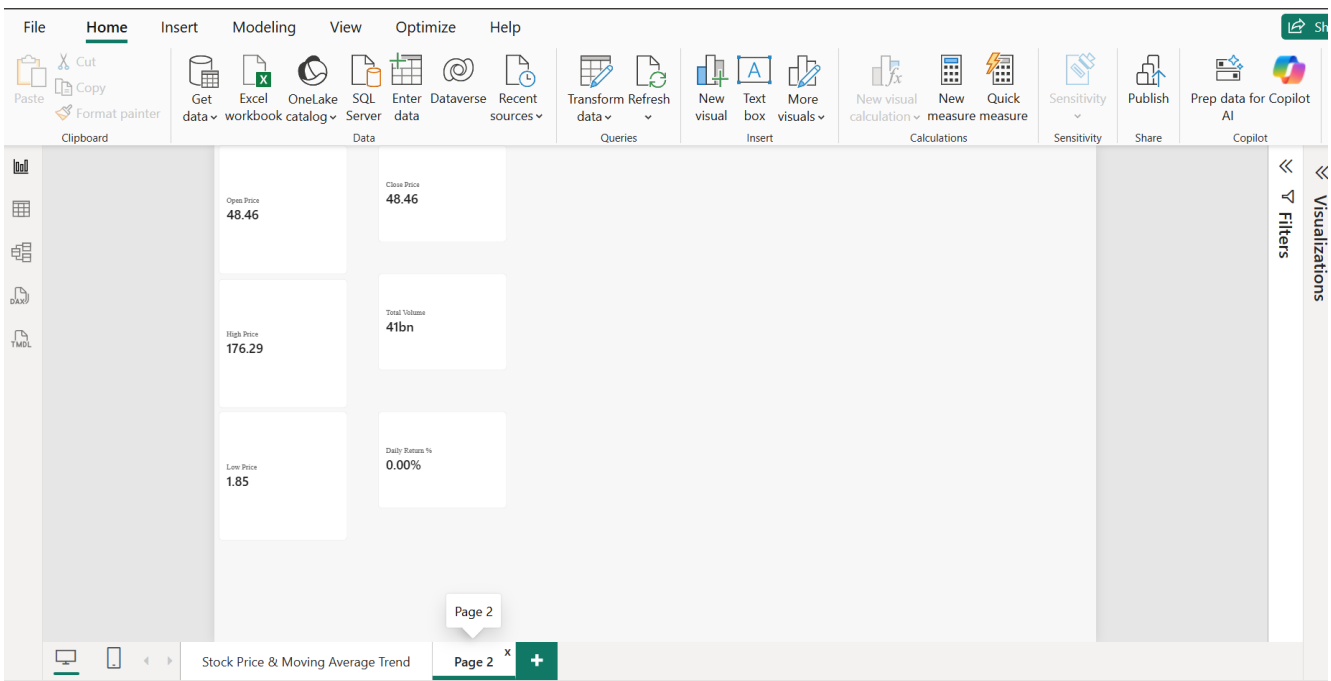


Figure 4. Tooltip-detail working page showing Open, High, Low, Close, Total Volume and Daily Return % cards.

Final configuration: Format page → Page information → Allow as tooltip = On; Canvas settings → Type = Tooltip. On the main chart, Format visual → General → Tooltips → Report page → Stock_Tooltip.

7. Final Dashboard Organization

Use the following structure when arranging the Power BI page. The purpose is to keep the main analytical charts large while moving controls into a clean side/bottom area.



Figure 5. Final layout reference generated for organizing the completed Week 4 dashboard. This is a visual reference, not a screenshot of Power BI.

Area	Recommended content
Header	SHOPIFY STOCK PERFORMANCE DASHBOARD + subtitle
Top KPI row	Average Close Price Highest Price Lowest Price Metric Selector
Main chart	Close Price + 20 SMA + 50 SMA + zoom slider
Second chart	Total Volume + Volume 20 SMA + spike highlighting
Controls	Between Date Range + Relative Date
Summary	Year / Month table if space allows
Tooltip	Stock_Tooltip page – not visible as a normal dashboard tile

8. Final Verification Checklist

Check	Status to verify before submission
20 SMA created	Measure exists and returns values
50 SMA created	Measure exists and returns values
Crossover signal	Uses the exact measure names created in the model
Price chart	Close + 20 SMA + 50 SMA visible
Continuous axis	X-axis Type = Continuous
Zoom slider	On
Volume chart	Total Volume + 20 SMA visible
Spike formatting	Conditional color applied to volume columns
Date Range	Between mode
Relative Date	Relative date mode configured
Metric Selector	Open / High / Low / Close available
Tooltip page	Stock_Tooltip + Allow as tooltip + Tooltip canvas
Tooltip metrics	Open, High, Low, Close, Volume, Daily Change %
Main chart tooltip	Points to Stock_Tooltip
Save project	Save the Week 4 .pbix file

GitHub submission structure

Recommended repository structure:

```
Shopify-Stock-Analysis/  
■ ■ ■ Week 3/  
■ ■ ■ ■ Week 3 Dashboard.pdf  
■ ■ ■ ■ Week 3 Report.pdf  
■ ■ ■ Week 4/  
■ ■ ■ ■ Shopify_Stock_Analysis_Week4.pbix  
■ ■ ■ ■ Shopify_Stock_Week4_Report.pdf  
■ ■ ■ ■ Week 4 Dashboard.pdf  
■ ■ ■ ■ README.md
```

Upload the Week 4 PDF report and dashboard PDF to the Week 4 folder. Upload the PBIX as well if its size is accepted by GitHub; GitHub's normal web upload has file-size limits, so a large PBIX may need another upload method.

Source note

This report follows the Week 4 task document supplied in the conversation. The screenshots are taken from the Power BI working views available in the project materials; the polished dashboard image is explicitly marked as a layout reference.